

CHANUTE, KS, USA

WMO: 724507

Lat: 37.670N

Lon: 95.484W

Elev: 985

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	12.6	-1.7	5.2	10.9	2.9	6.6	14.9	26.9	44.9	25.3	44.8	8.1	350	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	98.0	75.2	94.6	75.8	91.7	75.6	79.2	90.8	78.0	89.8	77.0	88.3	11.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.8	140.0	86.1	74.5	134.0	84.7	73.4	128.7	83.2	43.4	91.0	42.2	90.0	41.0	88.3	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	23.0	20.0	DB	-0.4	100.9	6.3	4.2	-4.9	103.9	-8.6	106.4	-12.2	108.8	-16.8	111.9	(4)
(5)			WB	-1.0	81.0	6.0	1.2	-5.3	81.8	-8.8	82.5	-12.2	83.2	-16.5	84.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	33.4	37.5	47.5	57.5	66.4	75.3	80.0	78.9	70.9	58.6	46.8	36.1	(6)
(7)		DBStd	18.55	11.05	11.19	10.80	9.02	7.59	5.65	5.33	5.65	7.50	9.12	9.91	10.26	(7)
(8)		HDD50	1750	523	373	175	35	2	0	0	0	0	27	173	442	(8)
(9)		HDD65	4347	980	771	547	255	78	5	0	1	28	238	548	896	(9)
(10)		CDD50	4490	8	22	99	258	510	760	930	895	626	294	77	11	(10)
(11)		CDD65	1611	0	0	5	28	122	314	465	431	204	40	2	0	(11)
(12)		CDH74	16609	0	5	46	253	1004	3112	5233	4639	1963	335	19	0	(12)
(13)		CDH80	7314	0	0	5	43	258	1230	2602	2308	791	76	1	0	(13)
(14)	Wind	WSAvg	9.5	9.7	10.2	11.1	11.6	9.7	9.2	7.8	7.6	8.0	9.2	9.8	9.7	(14)
(15)	Precipitation	PrecAvg	38.9	1.5	1.1	3.1	3.6	5.0	5.3	3.8	4.1	3.9	3.7	2.7	1.7	(15)
(16)		PrecMax	61.4	5.3	2.5	12.1	8.3	16.8	9.7	12.2	8.5	9.4	8.0	6.6	4.8	(16)
(17)		PrecMin	21.3	0.1	0.1	0.3	0.5	1.8	0.6	0.0	0.9	0.0	0.1	0.1	0.1	(17)
(18)		PrecStd	9.8	1.3	0.7	2.5	2.2	3.1	2.4	3.1	2.4	2.9	2.3	2.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.5	72.9	79.1	84.9	88.9	95.5	101.8	102.1	97.5	86.3	76.8	67.1	(19)
(20)			MCWB	55.2	58.2	63.5	66.7	71.7	77.6	74.8	74.2	73.1	70.6	62.2	58.2	(20)
(21)		2%	DB	60.9	66.0	73.8	80.2	85.6	92.0	98.2	98.3	91.4	81.8	71.2	61.4	(21)
(22)			MCWB	50.3	55.8	60.2	64.8	71.2	76.0	75.6	74.7	74.4	66.7	59.1	53.5	(22)
(23)		5%	DB	55.9	61.0	69.5	76.7	83.0	89.5	94.7	94.8	88.1	77.6	67.1	56.1	(23)
(24)			MCWB	47.8	52.1	57.5	62.9	70.5	75.6	76.6	75.3	73.0	64.5	57.7	49.6	(24)
(25)		10%	DB	50.2	55.4	65.4	73.0	79.9	87.0	91.7	91.5	84.4	73.6	62.7	51.9	(25)
(26)			MCWB	42.8	47.9	56.0	61.4	69.0	74.1	76.5	74.8	70.5	64.0	54.4	45.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	61.9	66.2	70.2	75.7	80.1	81.1	80.2	77.8	73.4	65.4	61.8	(27)
(28)			MCD B	64.0	68.1	76.1	80.5	84.9	91.3	91.9	91.8	89.2	81.5	72.1	65.8	(28)
(29)		2%	WB	53.3	57.6	63.0	67.8	74.0	78.2	79.4	78.5	76.2	71.0	62.1	55.6	(29)
(30)			MCD B	59.0	64.5	70.3	76.0	82.3	89.1	90.8	90.8	87.3	77.7	68.5	59.6	(30)
(31)		5%	WB	47.7	53.0	60.1	65.6	72.3	76.9	78.3	77.2	74.6	68.3	59.2	50.7	(31)
(32)			MCD B	55.3	59.6	66.9	73.5	80.2	87.1	90.0	89.7	84.8	74.1	65.3	54.8	(32)
(33)		10%	WB	43.5	47.6	56.6	63.2	70.4	75.4	77.3	75.8	72.9	65.2	55.7	46.1	(33)
(34)			MCD B	50.9	55.3	63.9	71.4	77.9	84.6	89.0	87.1	82.0	72.0	61.6	51.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.7	20.2	21.7	21.6	19.9	19.4	20.3	21.2	21.9	21.8	21.2	19.0	(35)
(36)			MCD BR	28.4	28.4	27.6	25.7	21.9	21.2	23.7	25.3	24.3	25.5	25.8	25.6	(36)
(37)			MCWBR	18.8	18.8	16.2	14.6	11.2	9.5	7.6	8.0	10.1	13.1	15.8	17.9	(37)
(38)		5% WB	MCD BR	24.8	25.4	21.9	21.9	18.9	19.4	20.1	20.9	20.8	19.4	21.9	21.3	(38)
(39)			MCWBR	17.9	18.2	14.9	14.4	11.1	9.9	8.3	8.7	9.4	12.3	15.7	17.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.296	0.307	0.343	0.380	0.417	0.423	0.435	0.439	0.384	0.344	0.308	0.288	(40)	
(41)		taud	2.471	2.454	2.398	2.318	2.275	2.279	2.293	2.263	2.377	2.469	2.512	2.517	(41)	
(42)		Ebn at Noon	284	294	292	285	275	272	268	265	274	277	277	280	(42)	
(43)		Edh at Noon	26	30	35	40	43	42	42	42	35	29	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	759	977	1277	1619	1813	2082	2073	1855	1549	1136	817	648	(44)	
(45)		RadStd	73	150	84	110	167	172	138	114	117	136	89	64	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page